Self-Information (Surprisal)

math-foundations / 35-self-information

1 Motivation

Information theory begins with a deceptively simple question: how much *information* does the occurrence of a single event convey? Intuition gives three desiderata.

- A certain event ($p = 1$) conveys no information.
- Rarer events convey more information than common ones.
- Two independent events convey, jointly, the sum of their individual informations.

Shannon (1948) showed that these three properties, together with continuity, single out a one-parameter family of functions: $I(p) = -c \log p$. The constant $c > 0$ merely chooses units; up to that choice, $-\log p$ is the *unique* measure of surprisal.

2 Definitions

Definition 1 (Self-information / surprisal)**.** Let X be a discrete random variable with probability mass function p on a countable alphabet $\mathcal{X}$, and let $x \in \mathcal{X}$ satisfy $p(x) > 0$. The *self-information* of the outcome x is

$$I(x) \;=\; -\log p(x).$$

The base of the logarithm fixes the units of I:

$$\log_2 \;\Rightarrow\; \text{bits}, \qquad \ln \;\Rightarrow\; \text{nats}, \qquad \log_{10} \;\Rightarrow\; \text{hartleys}.$$

Definition 2 (Self-information of an event)**.** For an event A with $\Pr(A) > 0$, define $I(A) = -\log \Pr(A)$. When $A = \{X = x\}$ this recovers the previous definition.

Remark 3. Self-information is a function of p, not of x in any geometric sense; two outcomes with the same probability carry the same surprisal regardless of their identity.

3 Additivity for independent events

Theorem 4 (Additivity)**.** *Let A and B be independent events with probabilities $p_A, p_B \in (0, 1]$. Then*

$$I(A \cap B) \;=\; I(A) + I(B).$$

Proof. By the definition of independence,

$$\Pr(A \cap B) = p_A p_B.$$

Applying the definition of self-information,

$$I(A \cap B) = -\log \Pr(A \cap B) = -\log(p_A p_B).$$

Using the homomorphism property of the logarithm, $\log(xy) = \log x + \log y$ for $x, y > 0$, we obtain

$$-\log(p_A p_B) = -\log p_A - \log p_B = I(A) + I(B). \qquad \square$$

Remark 5. Independence is essential. If $B = A$, then $A \cap B = A$ and $I(A \cap B) = I(A) \neq 2I(A)$ in general. The correct generalisation to dependent events introduces *conditional* self-information $I(B \mid A) = -\log \Pr(B \mid A)$, yielding the chain rule $I(A \cap B) = I(A) + I(B \mid A)$.

4 Uniqueness of $-\log p$

The additivity property is, in fact, almost sufficient to characterise surprisal.

Theorem 6 (Shannon uniqueness, abridged)**.** *Let $f : (0, 1] \to \mathbb{R}_{\geq 0}$ be continuous, strictly decreasing, and satisfy*

$$f(pq) = f(p) + f(q) \qquad \text{for all } p, q \in (0, 1].$$

Then there exists a constant $c > 0$ such that $f(p) = -c \log p$.

Sketch. Set $g(t) = f(e^{-t})$ for $t \in [0, \infty)$. The functional equation becomes

$$g(s + t) = f(e^{-(s+t)}) = f(e^{-s}e^{-t}) = f(e^{-s}) + f(e^{-t}) = g(s) + g(t),$$

i.e. g is additive on $[0, \infty)$. Together with continuity, Cauchy's functional equation gives $g(t) = ct$ for some constant c. Strict monotonicity of f forces $c > 0$. Translating back, $f(p) = g(-\log p) = -c \log p$. $\square$

5 Worked example

Example 7 (Biased coin)**.** Let $X \in \{H, T\}$ with $\Pr(X = H) = 0.9$, $\Pr(X = T) = 0.1$. Working in bits:

$$I(H) = -\log_2(0.9) \approx 0.152, \qquad I(T) = -\log_2(0.1) \approx 3.322.$$

The expected self-information is

$$\mathbb{E}[I(X)] = 0.9 \cdot 0.152 + 0.1 \cdot 3.322 \approx 0.469 \text{ bits},$$

the Shannon entropy of X, which is strictly less than the maximum of 1 bit achieved by a fair coin.

6 Connection to machine learning

For a language model $q(x_t \mid x_{<t})$, the per-token cross-entropy loss

$$\mathcal{L}_t = -\log q(x_t \mid x_{<t})$$

is precisely the self-information that the model assigns to the observed token. Minimising average loss is therefore minimising mean surprisal under the model. Perplexity, $\exp(\overline{\mathcal{L}})$, is the exponentiated mean surprisal — the geometric-mean inverse-probability per token.